# Investigating the use of Reinforcement Learning Algorithms in Traffic Evnrionments

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[1]**Disclaimer:** This report is submitted as part requirement for the BSc Degree in Computer Science at UCL. It is substantially the result of my own work except where explicitly indicated in the text. The report may be freely copied and distributed provided the source is explicitly acknowledged.

**Abstract**

Summarise your report concisely.

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# Chapter 1

# Title of first chapter

This is just a bare minimum to get started. There is unlimited guidance on using latex, e.g., `https://en.wikibooks.org/wiki/LaTeX`. You are still responsible to check the detailed requirements of a project, including formatting instructions, see `https://moodle.ucl.ac.uk/pluginfile.php/3591429/mod_resource/content/7/UGProjects2017.pdf`. Leave at least a line of white space when you want to start a new paragraph.

## 1.1 Section 1

Chapters should contain numbered sections and sub-sections.

### 1.1.1 Sub-Section 1

Remember that this is a structured technical/academic report, not an essay.

## 1.2 Mathematical Notation

Mathematical expressions are placed inline between dollar signs, e.g. $\sqrt{2}, \sum_{i=0}^{n} f(i)$, or in display mode

$$e^{i\pi} = -1$$

and another way, this time with labels,

$$A = B \wedge B = C \rightarrow A = C \tag{1.1}$$

$$\rightarrow C = A \tag{1.2}$$

note that

$$n! = \prod_{1 \leq i \leq n} i \tag{1.3}$$

$$\int_{x=1}^{y} \frac{1}{x} \mathrm{d}x = \log y \tag{1.4}$$

We can refer to labels like this (1.1).

# Chapter 2

# Title of second chapter

The main chapters should start with a brief summary – this chapter is about, or introduces, some aspect of the work.

## 2.1 Main Content

Often lots of citations here (and elsewhere), e.g. [Rey97] or [Pri70, Theorem 2.3]. Bibtex can help with this, but is not essential. If you want pictures, try

aristotle.jpg

You can use

- lists
- like this

or numbered

1. like this,
2. or this

but don't overdo it.

## 2.2 More Content

## 2.3 Summary

The main chapters should have a short summary, highlighting what the chapter has covered and any key points that need emphasising.

# Chapter 3

# Title of third chapter

If you have a formal theorem you might try this.

**DEFINITION 1** *See definition 1.*

**THEOREM 2** *For all* $n \in \mathbb{N}$, $1^n = 1$.

PROOF:

By induction over $n$. $\square$

# Chapter 4

# etc.

# Chapter 5

# Conclusions

## 5.1 Achievements

Summarise the achievements to confirm the project goals have been met.

## 5.2 Evaluation

Evaluation of the work (this may be in a separate chapter if there is substantial evaluation).

## 5.3 Future Work

How the project might be continued, but don't give the impression you ran out of time!

# Bibliography

[Pri70] A. Prior. The notion of the present. *Studium Generale*, 23: 245–248, 1970.

[Rey97] M. Reynolds. A decidable temporal logic of parallelism. *Notre Dame Journal of Formal Logic*, 38(3): 419–436, 1997.

# Appendix A

# Other appendices, e.g., code listing

Put your appendix sections here